Test Plan Document

Vehicle Parking Management and Tracking System

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Legend: UT = Unit Test, IT = Integration Test, ST = System Test. 6 test cases are provided per functional requirement (IR-FR-01 to IR-FR-12), covering valid, invalid, and boundary scenarios.

Test Case ID	Module	Description	Pre-conditions	Test Steps	Test Data	Expected Result	Actual Result	Result
IR-FR-01 — Register/Login								
UT_01	Register/Login	Verify registration with valid details	App accessible	1. Open register 2. Enter valid details 3. Submit	Name: Arjun, Email: valid, Pwd: Abc@123	Account created, confirmation shown	-	-
UT_02	Register/Login	Verify login with valid credentials	User is registered	1. Open login 2. Enter username/password 3. Click login	Registered username/password	User authenticated, dashboard shown	-	-
IT_01	Register/Login	Verify login rejects invalid password	User is registered	1. Enter correct username 2. Enter wrong password 3. Login	Valid username, wrong password	Error message displayed, access denied	-	-
IT_02	Register/Login	Verify registration rejects duplicate email	Email already registered	1. Enter existing email 2. Submit	Existing email address	Error: email already in use	-	-
IT_03	Register/Login	Verify registration rejects invalid email format	App accessible	1. Enter malformed email 2. Submit	Email: abc@com	Validation error shown, form not submitted	-	-
ST_01	Register/Login	Verify account locks after 5 failed login attempts	User is registered	1. Enter wrong password 5 times consecutively	Valid username, wrong password x5	Account temporarily locked (IR-SEC-04)	-	-
IR-FR-02 — Manage Vehicle								
UT_03	Manage Vehicle	Verify adding a new vehicle	User logged in	1. Open vehicle management 2. Enter vehicle details 3. Save	Reg No: KA01AB1234, Type: Car	Vehicle saved and displayed	-	-
UT_04	Manage Vehicle	Verify viewing registered vehicle list	User has registered vehicles	1. Open vehicle management screen	N/A	List of user's vehicles displayed	-	-
IT_02	Manage Vehicle	Verify duplicate vehicle registration is rejected	Vehicle already registered	1. Enter same reg number 2. Submit	Existing Reg No: KA01AB1234	Error: vehicle already exists	-	-
IT_04	Manage Vehicle	Verify vehicle registration rejects invalid reg number format	User logged in	1. Enter invalid reg number 2. Submit	Reg No: 1234	Validation error shown	-	-
IT_05	Manage Vehicle	Verify updating vehicle details	Vehicle exists	1. Select vehicle 2. Edit color 3. Save	Color: Red -> Blue	Updated details saved	-	-
ST_02	Manage Vehicle	Verify updated vehicle info persists after logout/login	Vehicle exists	1. Update vehicle 2. Logout 3. Login 4. View vehicle	Color: Blue	Updated color displayed correctly	-	-
IR-FR-03 — Parking Availability								
UT_05	Parking Availability	Verify slot status displayed correctly	Slots exist in system	1. Open availability screen	N/A	Slots shown as Available/Occupied/Reserved/Unavailable	-	-

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UT_06	Parking Availability	Verify only available slots shown when filter applied	Mixed slot statuses exist	1. Open availability 2. Apply 'Available' filter	Filter: Available	Only available slots listed	-	-
IT_06	Parking Availability	Verify availability updates after a slot is occupied	A slot is assigned to a vehicle	1. Assign slot 2. Refresh availability screen	Slot ID: P-12	Slot P-12 shown as Occupied	-	-
IT_07	Parking Availability	Verify availability updates after a slot is released	Vehicle exits	1. Process exit 2. Refresh availability screen	Slot ID: P-12	Slot P-12 shown as Available	-	-
IT_08	Parking Availability	Verify attendant can view availability alongside driver	Attendant logged in	1. Open availability as attendant	N/A	Same real-time slot data displayed	-	-
ST_03	Parking Availability	Verify availability screen loads within performance limit	Normal load conditions	1. Open availability screen 2. Measure response time	N/A	Loads within 2 seconds for 95% requests (IR-NFR-01)	-	-
IR-FR-04 — Reserve Slot								
UT_07	Reserve Slot	Verify reservation of an available slot	User logged in, slot available	1. View slots 2. Select slot 3. Choose time period 4. Confirm	Slot: P-05, 2-4 PM	Reservation created, slot marked Reserved	-	-
UT_08	Reserve Slot	Verify reservation requires login	User not logged in	1. Attempt to reserve a slot	Slot: P-05	User redirected to login	-	-
IT_09	Reserve Slot	Verify reservation fails for already reserved slot	Slot already reserved	1. Select reserved slot 2. Confirm	Slot: P-05 (reserved)	Error shown, alternate slot suggested	-	-
IT_10	Reserve Slot	Verify reservation rejects invalid time period	Slot available	1. Select slot 2. Enter end time before start time	Start 4 PM, End 2 PM	Validation error shown	-	-
IT_11	Reserve Slot	Verify reserved slot cannot be assigned to another vehicle during reservation window	Slot reserved for 2-4 PM	1. Attempt entry using reserved slot at 3 PM for different vehicle	Slot: P-05, different vehicle	Assignment blocked (DR-08)	-	-
ST_04	Reserve Slot	Verify reservation record reflected in admin occupancy report	Reservation created	1. Create reservation 2. Open admin occupancy view	Slot: P-05	Occupancy count reflects the reservation	-	-
IR-FR-05 — Vehicle Entry								
UT_09	Vehicle Entry	Verify recording of vehicle entry	Attendant logged in, slot available	1. Enter vehicle number 2. Confirm entry	Reg No: KA01AB1234	Session created, entry time recorded	-	-
UT_10	Vehicle Entry	Verify entry rejected for unregistered vehicle	Vehicle not registered	1. Enter unregistered vehicle number 2. Confirm	Reg No: KA99ZZ9999	Error: vehicle not found	-	-
IT_12	Vehicle Entry	Verify entry blocked if vehicle already has active session	Vehicle has active session	1. Attempt entry for same vehicle	Reg No: KA01AB1234 (active)	Error: active session already exists (DR-02)	-	-
IT_13	Vehicle Entry	Verify entry blocked when no slot is available	No slots available	1. Attempt vehicle entry	Reg No: KA01AB1234, no free slot	Error: no available slot	-	-
IT_14	Vehicle Entry	Verify entry records correct timestamp	Attendant logged in	1. Process entry 2. Check recorded entry time	Reg No: KA01AB1234	Entry time matches system clock	-	-
ST_05	Vehicle Entry	Verify entry transaction completes within time limit	Normal conditions	1. Perform entry 2. Measure duration	N/A	Completed within 3 seconds (IR-NFR-02)	-	-

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IR-FR-06 — Assign Slot								
UT_11	Assign Slot	Verify system assigns an available slot	Slot available	1. Process entry 2. System assigns slot	N/A	Suitable available slot assigned	-	-
UT_12	Assign Slot	Verify compatible slot assigned by vehicle type	Vehicle type: Two-wheeler	1. Process entry for two-wheeler	Vehicle type: Two-wheeler	Two-wheeler-compatible slot assigned (DR-07)	-	-
IT_15	Assign Slot	Verify occupied slot cannot be reassigned	Slot already occupied	1. Attempt to assign occupied slot	Slot: P-08 (occupied)	Assignment rejected (DR-03)	-	-
IT_16	Assign Slot	Verify unavailable slot cannot be assigned	Slot marked unavailable	1. Attempt to assign unavailable slot	Slot: P-15 (unavailable)	Assignment rejected	-	-
IT_17	Assign Slot	Verify manual slot override by attendant	Multiple slots available	1. Attendant manually selects preferred slot	Slot: P-20	Selected slot assigned instead of default	-	-
ST_06	Assign Slot	Verify slot status updates to Occupied system-wide	Slot assigned	1. Assign slot 2. Check availability screen	Slot: P-08	Status shows Occupied everywhere	-	-
IR-FR-07 — Track Vehicle								
UT_13	Track Vehicle	Verify searching a parked vehicle returns its slot	Vehicle has active session	1. Enter vehicle number 2. Search	Reg No: KA01AB1234	Current slot and status displayed	-	-
UT_14	Track Vehicle	Verify track vehicle accessible to administrator	Admin logged in	1. Admin searches vehicle	Reg No: KA01AB1234	Vehicle location displayed to admin	-	-
IT_18	Track Vehicle	Verify search for non-parked vehicle returns no active session	Vehicle not parked	1. Enter unparked vehicle number 2. Search	Reg No: KA09ZZ0000	Message: no active session found	-	-
IT_19	Track Vehicle	Verify search with invalid/empty vehicle number is handled	Attendant logged in	1. Leave field empty 2. Search	Reg No: (blank)	Validation message shown	-	-
IT_20	Track Vehicle	Verify track vehicle denied for unauthorized (driver) role	Driver logged in	1. Attempt vehicle search as driver	Reg No: any	Access denied (IR-SEC-02)	-	-
ST_07	Track Vehicle	Verify tracking result reflects latest slot after reassignment	Vehicle reassigned to new slot	1. Reassign slot 2. Search vehicle	Reg No: KA01AB1234	Updated slot shown, not stale data	-	-
IR-FR-08 — Vehicle Exit								
UT_15	Vehicle Exit	Verify recording of vehicle exit	Vehicle has active session	1. Enter vehicle number 2. Confirm exit	Reg No: KA01AB1234	Exit time recorded, session completed	-	-
UT_16	Vehicle Exit	Verify slot released and marked Available after exit	Exit processed	1. Process exit 2. Check slot status	Slot: P-08	Slot status changes to Available (SYS-12)	-	-
IT_21	Vehicle Exit	Verify exit is blocked without a prior active session	No active session	1. Attempt exit for unparked vehicle	Reg No: KA09ZZ0000	Error: no active session (DR-04)	-	-
IT_22	Vehicle Exit	Verify exit rejects exit time earlier than entry time	Active session exists	1. Attempt to set exit time before entry time	Entry 5 PM, Exit 3 PM	Validation error shown (DR-09)	-	-
IT_23	Vehicle Exit	Verify exit triggers fee calculation (includes UC-09)	Active session exists	1. Process exit 2. Check displayed fee	Entry 10:00, Exit 12:00	Fee calculated and displayed before completion	-	-
ST_08	Vehicle Exit	Verify completed session cannot be modified afterward	Session completed	1. Attempt to edit a completed session	Session ID: S-1001	Edit blocked (DR-10)	-	-
IR-FR-09 — Calculate Fee								

Test Case ID	Module	Description	Pre-conditions	Test Steps	Test Data	Expected Result	Actual Result	Result
UT_17	Calculate Fee	Verify fee computed correctly for given duration	Entry and exit times available	1. Process exit 2. System calculates fee	Entry 10:00, Exit 12:00, Rate ₹20/hr	Fee = ₹40 displayed	-	-
UT_18	Calculate Fee	Verify fee calculation for partial-hour duration	Entry/exit times available	1. Process exit with partial hour	Entry 10:00, Exit 10:45, Rate ₹20/hr	Fee rounded per configured rule and displayed	-	-
IT_24	Calculate Fee	Verify fee calculation uses correct configured rate	Rate configured by admin	1. Set rate 2. Process exit	Rate ₹30/hr, Duration 1.5 hr	Fee = ₹45 displayed	-	-
IT_25	Calculate Fee	Verify fee calculation handles zero-duration edge case	Entry and exit at same time	1. Process exit immediately after entry	Entry = Exit time	Minimum applicable fee applied, no error	-	-
IT_26	Calculate Fee	Verify fee updates correctly when admin changes rate mid-operation	New rate configured	1. Update rate 2. Process a new exit	New rate ₹25/hr	New fee calculated using updated rate	-	-
ST_09	Calculate Fee	Verify fee calculation completes without noticeable delay at exit	Normal load	1. Process exit 2. Measure time to display fee	N/A	Fee displayed as part of exit flow, within 3s (IR-NFR-02)	-	-
IR-FR-10 — Manage Payment								
UT_19	Manage Payment	Verify successful payment updates session status	Fee generated	1. Select payment method 2. Submit payment	Amount ₹40, Method: UPI	Payment recorded, status Paid	-	-
UT_20	Manage Payment	Verify payment options are displayed correctly	Fee generated	1. Open payment screen	N/A	Available payment methods listed	-	-
IT_27	Manage Payment	Verify failed payment is recorded correctly	Payment gateway available	1. Submit payment 2. Gateway declines	Amount ₹40, invalid card	Status: Failed, user notified	-	-
IT_28	Manage Payment	Verify payment rejects invalid/expired card details	Payment gateway available	1. Enter expired card 2. Submit	Expired card number	Validation/decline error shown	-	-
IT_29	Manage Payment	Verify attendant can process payment on behalf of driver	Attendant logged in	1. Attendant selects payment method 2. Confirms payment	Amount ₹40, cash	Payment recorded against session	-	-
ST_10	Manage Payment	Verify payment linked correctly to parking session	Payment completed	1. Complete payment 2. View session record	Session ID: S-1001	Payment associated with correct session (SYS-13)	-	-
IR-FR-11 — Parking History								
UT_21	Parking History	Verify history displays completed sessions	User has past sessions	1. Open parking history	User account with history	List of past sessions with fee/status shown	-	-
UT_22	Parking History	Verify history shows correct entry/exit/duration/fee fields	Completed session exists	1. Open history 2. Inspect a session record	Session ID: S-1001	Vehicle no., slot, entry/exit, duration, fee, status shown	-	-
IT_30	Parking History	Verify history record is not editable by normal user	Completed session exists	1. Attempt to edit past session	Session ID: S-1001	Edit action disabled/blocked (DR-10)	-	-
IT_31	Parking History	Verify empty history state for new user	New user, no sessions	1. Open parking history	New user account	Empty state / no-records message shown	-	-
IT_32	Parking History	Verify admin can view any user's parking history	Admin logged in	1. Admin opens a user's history	User: Driver X	Correct history for selected user displayed	-	-
ST_11	Parking History	Verify history list loads correctly for a user with many past sessions	User has 50+ sessions	1. Open parking history 2. Scroll/paginate	User with 50+ sessions	All records accessible without errors	-	-

Test Case ID	Module	Description	Pre-conditions	Test Steps	Test Data	Expected Result	Actual Result	Result
IR-FR-12 — Administration & Reports								
UT_23	Administration & Reports	Verify admin can generate an occupancy report	Admin logged in	1. Open reports 2. Select occupancy report 3. Generate	Date range: today	Report generated with current occupancy	-	-
UT_24	Administration & Reports	Verify admin can manage (add/remove) users	Admin logged in	1. Open user management 2. Add/remove a user	New user details	User list updated accordingly	-	-
IT_33	Administration & Reports	Verify only admin role can access audit logs	Non-admin user logged in	1. Attempt to open audit logs as attendant	Role: Attendant	Access denied (IR-SEC-08)	-	-
IT_34	Administration & Reports	Verify admin can update parking rate and it reflects in fee calc	Admin logged in	1. Update rate 2. Process a new exit	New rate ₹25/hr	New fee calculated using updated rate	-	-
IT_35	Administration & Reports	Verify admin can manage parking slots (add/disable a slot)	Admin logged in	1. Disable a slot 2. Check availability screen	Slot: P-30 disabled	Slot marked Unavailable, cannot be assigned	-	-
ST_12	Administration & Reports	Verify audit log records key administrative actions	Admin performs an action	1. Admin updates a rate 2. Open audit log	Action: rate update	Audit log shows action, admin, and timestamp (IR-SEC-07)	-	-